

RESEARCH ARTICLE

Multi-classification of brain tumor by using deep convolutional neural network model in magnetic resonance imaging images

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Abstract

Brain tumors are still diagnosed and classified based on the results of histopathological examinations of biopsy samples. The existing method requires extra effort from the user, takes too long, and can lead to blunders. These limitations underline the need of employing a fully automated deep learning system for the multi-classification of brain tumors. In order to facilitate early detection, this study employs a convolutional neural network (CNN) to multi-classify brain tumors. In this research, we present three distinct CNN models for use in three separate categorization tasks. The first CNN model can correctly categorize brain tumors 99.74% of the time. The second CNN model is 96.27% accurate in differentiating between normal, glioma, meningioma, pituitary, and metastatic brain tumors. The third CNN model successfully distinguishes between Grades II, III, and IV brain tumors 99.18% of the time. The Hybrid Particle Swarm Grey Wolf Optimization (HPSGWO) technique is used to quickly and accurately determine optimal values for all of CNN models most important hyperparameters. An HPSGWO algorithm is used to fine-tune all the necessary hyperparameters for optimal classification performance. The results are compared with standard existing CNN models across a range of performance measures. The proposed models are trained using publicly available large clinical datasets. To verify their initial multi-classification of brain tumors, clinicians and radiologists might use the proposed CNN models.

KEYWORDS

brain tumor, classification, convolutional neural network, deep learning, HPSGWO

1 | INTRODUCTION

A brain tumor is an abnormal growth of brain cells caused by the breakdown of the brain's regulatory mechanisms. Skull-based tumors have the potential to metastasize, push on the brain, and have detrimental effects on health. Medical imaging studies aimed at identifying and

categorizing brain tumors at an early stage are an active topic of study. This data is utilized to help doctors choose the best treatment options for their patients, which might ultimately save their lives. Brain tumors can be placed in any one of several categories. One common method of organization is the differentiation between benign and malignant brain tumors. Benign brain tumors develop in

the skull rather than the brain itself. Meningiomas are a crucial component of this group. Tumors in the brain are more likely to result in fatal complications than tumors in other organs. Meningiomas are an example of a benign tumor that, in very unusual circumstances, may develop into a malignant one. Since they seldom invade nearby brain structures, they are potentially curable by surgery. Pituitary tumors are abnormal growths in the glands that produce hormones and regulate other bodily processes. Pituitary tumors, on the other hand, are completely localized and harmless. Pituitary tumors are mostly harmless, however, they can develop into a malignancy. Damage to the pituitary gland can cause long-term hormone deficiencies and even blindness in extreme cases. Cancerous tumor cells are abnormal in that they divide and multiply in an unregulated manner.

The healthy tissues around them can be squished, invaded, or even killed by these tumors. Metastatic brain tumors are those that begin in another part of the body and metastasize to the brain. They often show up in the lungs, breasts, digestive system, skin, or prostate. The majority of malignant brain tumors are gliomas. Most brain tumors originate from these abnormally dividing cells. Rapid growth and infiltration into neighboring healthy tissues characterize these tumors, notwithstanding their infrequent ability to spread to the spinal cord or any other human organ.

The severity of gliomas allows for further categorization. The glioma tumor grading system developed by the World Health Organization (WHO) has become the standard in recent years. Gliomas are graded from I (benign) through IV using this system (malignant). This categorization is founded on histological features and survival statistics. Grades I and II are considered benign, but grades III (anaplastic astrocytoma) and IV (glioblastoma multiforme) are considered "high grade" or "malignant." Grade I tumors are the least invasive because they seldom spread to neighboring tissues. They normally develop extremely slowly but can be surgically corrected if necessary. Grade II brain tumors, unlike grade I tumors, tend to invade neighboring tissues and progress to more malignant forms. Grade III brain tumors seem strange when seen under a microscope. They need medical interventions apart from surgery because of their potential to infect other brain areas. Grade IV tumors, which are the most rapidly progressing, typically call for the most aggressive treatment options.

Using deep transfer learning and GoogleNet,¹ tackles a 3-class classification problem to differentiate between glioma, meningioma, and pituitary tumors, the three most frequent forms of brain tumors. When there aren't enough high-quality medical photos to go around, it's been found that transfer learning comes in handy. The

tumor detection system in² is a hybrid convolutional neural network (CNN). The framework is based on the Resnet50 concept. The last five levels of the Resnet50 model were eliminated, and eight more layers were added in their stead.

For the purpose of classifying brain tumors, a thorough survey is provided.³ With the use of deep learning technology, radiologists can predict tumor sites and further categorize them. It comprehensively discussed the benefits and drawbacks of every method for classifying brain tumors using deep learning. It then went on to give specifics on the available public datasets and the tools that may be used with them. Finally, this study outlined key problems, such as a dearth of publicly available datasets and comprehensive deep learning models, and offered concrete suggestions for future work in the field of brain tumor classification, such as exploring edge/fog/cloud computing with Federated Learning, modern data-enrichment techniques, model confidence and explainability, the Internet of Medical Things, and a comprehensive examination of sequential and transfer learning. Brain tumor classification methods may develop further as a result of being more suited for commercial clinical applications and being easily integrated with smart healthcare.

In,⁴ the authors provide a novel deep learning-based method for grading brain tumors. To classify brain tumors of varying grades, a CNN is first used to extract the tumor areas from the dataset, which is then augmented with additional parameters to increase the number of data samples, and ultimately, a pre-trained VGG-19 CNN model is trained and tuned. In,⁵ a CNN and genetic algorithm (GA)-based approach is created for noninvasively identifying glioma grades using magnetic resonance imaging (MRI). The suggested method employs GA to generate the architecture of the CNN, which is in stark contrast to the present approaches, which are often dependent on trial and error or on adopting fixed common patterns. In addition, GA's best model is utilized with bagging, an ensemble method, to reduce the prediction error variance.

For the purpose of using computers to help in the diagnosis of brain tumors, a random forest classifier is designed.^{6,7} The recovered gray-level co-occurrence matrix from the preprocessed brain picture is used to identify and reduce the noise present in the MRI of the brain. In,⁸ a hybrid intelligent approach is proposed for MRI-based brain tumor categorization. Features are extracted, dimensions are reduced, and then the data is classified. In the first phase, they extracted MRI image features using discrete wavelet modification. Second, principal component analysis (PCA) was used to identify the most relevant features of MRI scans. For the purpose

of classification, two classifiers have been developed. The first classifier is based on an FP-ANN (feed-forward back-propagation ANN), while the second is based on the k-nearest neighbor algorithm. (k-NN).

For the classification of MR images of brain tumors, the authors of⁹ describe a neural dictionary learning with a local constraint approach. It employs a multi-layer dictionary learning architecture within a CNN framework to probe the discriminative information. The sparse coding vector that results from projecting a vector onto a dictionary into many separate spaces is intuitive. Meanwhile, they have developed a supervised k-nearest neighbor network to act as a local restriction on atoms, resulting in a dictionary with high discriminatory power without sacrificing geometric precision. To solve the problem at hand, an efficient iterative optimization strategy is developed. An efficient technique for MRI image segmentation is created.¹⁰ First, a novel objective function that takes into account local spatial information and bias correction is developed using kernelized fuzzy entropy clustering. Second, Particle Swarm Optimization (PSO) algorithm is used with a modified fitness function to create an improved method for MRI picture segmentation.

The authors of^{11,12} described a computer-aided diagnosis system that uses a novel deep neural network structure to classify MR images of brain tumors into three broad categories: meningioma, glioma, and pituitary. Gliomas were then further classified into several grades: Grade II, Grade III, and Grade IV. The constructed network consists of 16 layers, the first of which stores the input pictures before processing, followed by three convolution layers, three ReLU levels, three normalization layers, and three max-pooling layers. In order to anticipate the output and generate the predicted class, a fully connected layer, a softmax layer, and a classification layer are employed. To further counteract potential over-turning, two dropout layers are employed.

For brain tumor diagnosis in MRI images, we create an Adaptive Neuro-Fuzzy Inference System (ANFIS).^{13,14} Tumor segmentation and edema have both been employed. Features of the segmented tumor parts are retrieved from the images using discrete wavelet transform, wavelet series transform, Edge, and a color histogram. The features used by the ANFIS system to determine whether a tumor is benign or malignant are selected using the Modified Grey Wolf Optimizer algorithm. The K-Means Clustering method performs the adjusting. In,¹⁵ a deep CNN with completely optimized frameworks is described for the multi-class classification of brain tumors. It categorized the various types and stages of tumors.

In,^{16,17} CNN techniques are used to accurately categorize brain tumor grades, and in,^{18,19} a variety of

classification strategies are compared and contrasted. For the purpose of classifying brain tumors in public datasets, a hybrid CNN is designed in.^{20–22} They demonstrated that by tuning the CNN's hyperparameters with a specialized approach, accuracy can be improved.

A SpCaNet (Spinal Convolution Attention Network) is proposed in²³ which employs a Positional Attention (PA) convolution block, a Relative self-attention transformer block, and an Intermittent Fully Connected (IFC) layer to make the most of the pathological characteristics of brain tumors. When it comes to detecting brain tumors, it is more portable and effective. In addition, they propose the gradient awareness minimization (GAM) strategy for training the SpCaNet model.

In,^{24,25} they proposed an optimized CNN for the multi-classification of brain tumors by using an improved dragonfly optimization algorithm and Harris Hawks algorithm, respectively. PatchResNet and PSPNet, namely pyramid scene parsing networks are proposed in^{26,27} for the multi-classification of brain tumors in different datasets. In,^{28,29} they have presented an optimized CNN model for the multi-classification of different diseases from multiple datasets.

Brain tumors must be correctly graded and categorized at an early stage for accurate diagnosis, therapy planning, and evaluation of treatment efficacy. Histological examination of biopsy specimens is still used for detecting, classifying, and grading brain cancers, despite advances in modern medical technology. Pathological tests, clinical evaluations, and the analysis of imaging studies like CT scans and MRIs are often used to arrive at the definitive diagnosis. It is common knowledge that this method of diagnosis has a number of limitations, chief among them being obtrusive, time-consuming, and prone to sample mistakes. Medical professionals, such as doctors and radiologists, can improve their diagnostic skills and reduce the time it takes to arrive at an accurate diagnosis with the use of computer-aided, completely automated detection and diagnosis systems.

Using publicly available data, in this paper we have developed three unique CNN models for the multi-classification of brain tumors. This is the first effort at using CNN to classify many kinds of brain tumors in MRI images, and the Hybrid Particle Swarm Grey Wolf Optimization (HPSGWO) optimizes almost all of CNNs' hyperparameters automatically. The newly proposed original CNN model for Classifications_1 work improves the accuracy of identifying brain tumors to 99.33%. Applying the recommended CNN model for Classifications_2 work on MRI images of glioma, meningioma, pituitary, metastatic tumor types, and healthy yields a classification accuracy of 97.27%. When applied to sorting brain tumor images into Grade II, Grade III, and Grade

IV categories, the CNN model for Classification_3 achieves an impressive classification accuracy of 99.18%.

The rest of the paper is arranged as follows: We provide Preliminaries in Section 2. The proposed Methodology and Experimental results are presented in Sections 3 and 4, respectively. Comparison and Discussion are placed in Section 5. In Section 6 we provide the Conclusion of the paper.

2 | PRELIMINERIES

2.1 | Convolutional neural networks

When it comes to neural networks, the CNN model is by far the most used deep learning model. The two main components of a regular CNN model are feature extraction and classification. The five main layers that make up a CNN architecture are the input layer, convolution layer, pooling layer, fully connected layer, and classification layer. CNN extracts feature and classifies data by stacking trainable layers in a certain order. Convolutional and pooling layers are commonly employed in the CNN's feature extraction stage, whereas fully connected and classification layers are commonly used in the CNN's classification step. CNNs have been widely utilized in various sectors and can analyze any signal, including audio and video, despite the recent emphasis on image categorization and the acceptance of pictures as input data.

A CNN's individual layers all have an important role. The size and number of kernels determine how many features are extracted from each layer. During model training, the kernel weights are randomized and then learned. The ReLu layer receives its input from the convolution layer's output. The function of nonlinear activation for the convolution layer, ReLu always uses the same stable value. Because of its ease of use and absence of negativity, CNN often employs the ReLU activation function. The dropout layer is applied after ReLu to prevent overfitting of the model. After that, the pooling layer helps to downsample the feature map. The representation's sensitivity to small changes in the input is improved as a result. The pooling layer takes each feature map and uses it to create a new set of weight maps. To get the most information out of each feature map patch, we use max pooling. It helps blur out the rougher parts of an image, including the edges and corners. Fully Connected (FC) layers are commonly used in the deeper levels of a CNN. The output of the pooling layer is used as input to the FC layers, which then do a classification. The softmax function is used to generate a multiclass

classification probability distribution in the last layer of the CNN model. To prevent overfitting, the CNN model employs regularization. Regularization helps lessen overfitting when a penalty is included in the loss function. The dropout strategy reduces interdependent learning to prevent overfitting in CNN. After a CNN's structure has been defined, the core weights are adjusted via backpropagation in order to address the problem at hand. In order to improve the CNN model's overall efficacy, it is helpful to think about hyper-parameter tuning as an optimization issue.

2.2 | Particle swarm optimization algorithm

The PSO algorithm was developed in part to mimic the social behavior of animals, such as fish schooling and bird assembling. The birds scatter or assemble in search of food before arriving at a spot where they may receive it. One bird can always smell the food quite well, even though the others are searching in other areas. This exhibits the bird's intelligence and hunting prowess by allowing it to communicate with its prey and locate potential meals. The birds will flock together as the messengers fly to the source of the food. Each individual in the swarm or crowd is referred to as a particle in this animal behavior-based method for addressing global optimization functions or issues. Two equations constantly adjust the coordinates of each crowd member inside the PSO algorithm's global search space. These are the related mathematical formulas:

$$v_y^{t+1} = w \times v_y^t + c_1 e_1 (lBest_y^t - x_y^t) + c_2 e_2 (gBest_y^t - x_y^t) \quad (1)$$

where v_y^{t+1} is the next velocity of each particle, the initial inertia is w , the previous velocity of particle y at time t is v_y^t , c_1 and c_2 are the random numbers ranges $[0,1]$. $lBest_y^t$ and $gBest_y^t$ are the local and global fitness function of the swarm, and the previous position of swarm y at time t is x_y^t .

$$lBest_y^{t+1} = \begin{cases} lBest_y^t & \text{iff } lBest_y^t \leq f(l_y^{t+1}) \\ l_y^{t+1} & \text{iff } lBest_y^t \geq f(l_y^{t+1}) \end{cases} \quad (2)$$

$$gBest_y^{t+1} = \max[f(y), f(gBest_y^t)] \quad (3)$$

where $y \in gBest_0^t, gBest_1^t, gBest_2^t, \dots, gBest_n^t$.

2.3 | Grey wolf optimization algorithm

Among optimization algorithms, the Grey Wolf Optimization (GWO) algorithm uses metaheuristics. Grey wolf social organization and hunting techniques served as models. The remaining subordinate wolves are categorized as gamma, with each pack having an alpha, beta, and omega wolf. The leadership hierarchy is simulated using the four groups that make up the GWO population: *alpha*, *beta*, *gamma*, and *omega*. All decisions within the pack are made by the alpha wolves, who are also in command of the hunt. The alpha wolf's subordinate beta wolves back up its assertions. Omega, who are situated in the following rung of the group, maintain the dominant hierarchy of the group. The other wolves are referred to as Gamma since they answer to the Omega. Based on fitness and optimality, GWO solutions are divided into three levels. Most likely, the selection of alpha is the best one for the optimization assignment. The optimization problem is resolved using swarm intelligence techniques when a leader is not available to supervise the group for the entire time period. The GWO approach avoids this problem by putting grey wolves in positions of leadership. Swarm intelligence improvisations made use of GWOs since they were modeled like grey wolves and could solve problems involving image recognition and categorization. Hunting, locating prey, assaulting prey, and surrounding prey are the key GWO activities involving optimization. The mathematical Equations (4) and (5) show the encircling behavior of each swarm agent.

$$\vec{D} = |\vec{E} \cdot \vec{Z}_{y(t)} - \vec{Z}_{(t)}| \quad (4)$$

$$\vec{Z}_{(t+1)} = |\vec{Z}_{y(t)} - \vec{B} \cdot \vec{D}| \quad (5)$$

where current iteration is t , position of the prey is Z_y , position of GreyWolf is Z , and B, E are coefficient vectors. D is the enrichment behavior of each agent. Equations (6) and (7) can be used to calculate B and D .

$$\vec{B} = b \times \vec{e}_1 - \vec{e}_1 \quad (6)$$

$$\vec{E} = 2 \times \vec{e}_2 \quad (7)$$

Here, b is linearly reduced from 2 to 0 as a result of a certain number of iterations using e_1 and e_2 whose values are chosen from a range of $[0, 1]$. The position of the GreyWolf in relation to the position of the prey is updated using Equations (8)–(10).

$$\vec{D}_\alpha = |\vec{E}_1 \cdot \vec{Z}_\alpha - \vec{Z}_{(t)}| \quad (8)$$

$$\vec{D}_\beta = |\vec{E}_2 \cdot \vec{Z}_\beta - \vec{Z}_{(t)}| \quad (9)$$

$$\vec{D}_\gamma = |\vec{E}_3 \cdot \vec{Z}_\gamma - \vec{Z}_{(t)}| \quad (10)$$

$$\vec{Z}_1 = |\vec{Z}_\alpha - A_1 \vec{D}_\alpha| \quad (11)$$

$$\vec{Z}_2 = |\vec{Z}_\beta - A_2 \vec{D}_\beta| \quad (12)$$

$$\vec{Z}_3 = |\vec{Z}_\gamma - A_3 \vec{D}_\gamma| \quad (13)$$

where $\vec{D}_\alpha, \vec{D}_\beta, \vec{D}_\gamma$ are the distance vectors and $\vec{Z}_1, \vec{Z}_2, \vec{Z}_3$ are the positions vector wolf. Grey Wolves positions are updated by:

$$\vec{Z}_{(t+1)} = \frac{\vec{Z}_1 + \vec{Z}_2 + \vec{Z}_3}{3} \quad (14)$$

3 | PROPOSED METHODOLOGY

Figure 1 shows the proposed workflow of the brain tumor classification system.

3.1 | Datasets

The research takes use of two publicly available datasets. The first dataset is referred to as RIDER.³⁰ It is a focused data collection that includes MRI-multi-sequence images from 19 glioblastoma patients (Grade IV). This collection has 70 220 images in total. REMBRANDT is the name of the second dataset.³¹ It consists of MRI images of 130 individuals with gliomas of varying grades (II, III, and IV). There are 1,10 020 images in total in this dataset. A total of 3990 images were used for the Classification_1 work in which there were 2140 tumors and 1850 tumor-free images. A total of 6450 images were used for the Classification_2 work which consists of 1350 normal, 1450 glioma, 1250 metastatic, 1200 meningioma, and 1200 pituitary images. During the Classification_3 work, 6070 images in total are used, including 2176 grade II, 1718 grade III, and grade III of 2176 images. Table 1 contains all the information regarding the datasets and Figure 2 shows an example of brain tumor images from the datasets.

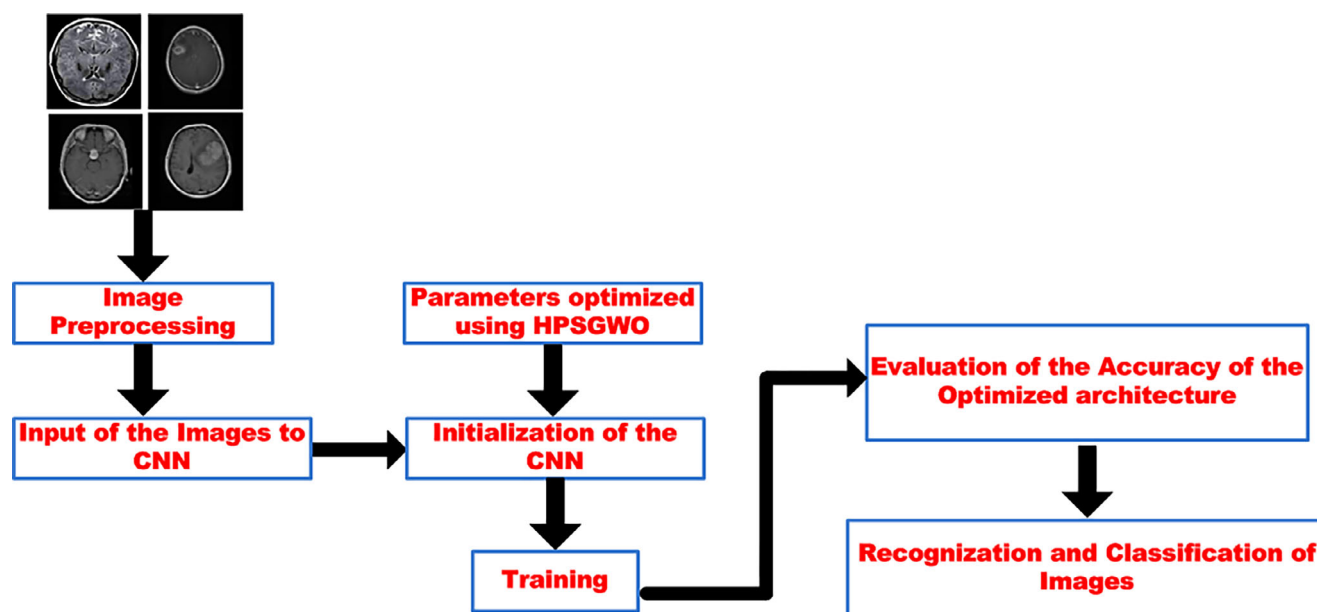


FIGURE 1 Proposed methodology for classification of brain tumors.

TABLE 1 Details of a dataset.

Classification Work	Classification class	Image count for each class	Sum of all images
Classification_1	Tumor	2140	3990
	No Tumor	1850	
Classification_2	Normal	1350	6450
	Glioma	1450	
	Meningioma	1200	
	Pituitary	1200	
	Metastatic	1250	
Classification_3	II	2176	6070
	III	1718	
	IV	2176	

3.2 | CNNs optimized with Hybrid Particle Swarm Grey Wolf Optimization algorithm

As the application of CNNs in the field of medical image processing has increased, a number of difficulties have arisen in their implementation. More computational costs are incurred when designs are refined to yield better outcomes and input images increase in quality. In order to reduce these processing costs and get better results, it is essential to use reliable hardware and fine-tune the hyperparameters of the current network. As a result, as shown in Figure 3, the suggested CNN models have had the HPSGWO approach automatically adjust their critical hyperparameters. The HPSGWO method outperforms conventional CNN hyperparameter optimizations when

working with a constrained search space. The HPSGWO seeks to identify the optimal value of the hyperparameters of a given network that has been trained. The HPSGWO has been developed without modifying the fundamental computations of either the PSO or the GWO methods. Using the PSO approach, almost all problems encountered in the real world may be fixed efficiently. However, there should be a technique to discover the local minimal probability of PSO fooling. In our suggested technique, the GWO is employed to help PSO avoid drifting toward a local minimum. Some particles are moved to random places with a little probability of success using the PSO algorithm in order to prevent local minima. To circumvent these risks, we apply the GWO exploration capacity to direct certain particles to areas that are marginally bolstered by the GWO approach. However, adding the GWO technique to the

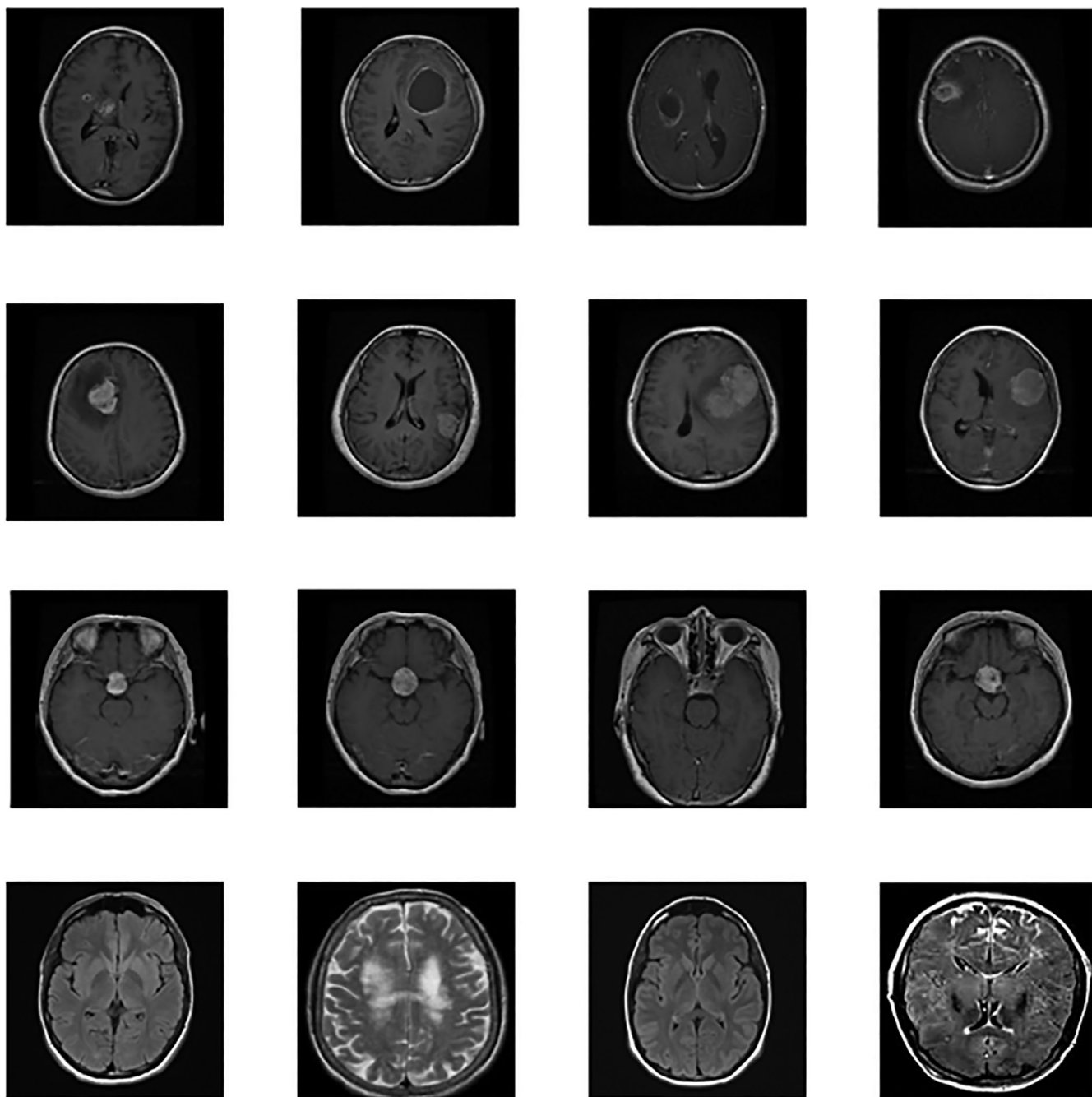


FIGURE 2 Examples of brain tumor samples from the datasets.

PSO algorithm increases the running time. The two features are combined in the HPSGWO algorithm:

$$\vec{D}_\alpha = |E_1 \cdot \vec{Z}_\alpha - w \times \vec{Z}_{(t)}| \quad (15)$$

$$\vec{D}_\beta = |E_2 \cdot \vec{Z}_\beta - w \times \vec{Z}_{(t)}| \quad (16)$$

$$\vec{D}_\gamma = |Z_3 \cdot \vec{Z}_\gamma - w \times \vec{Z}_{(t)}| \quad (17)$$

In order to update the velocities, Equations (18) and (19) integrate PSO and GWO variations:

$$v_y^{(t+1)} = w \times \left(v_y^t + c_1 e_1 (x_1 - x_y t) + c_2 e_2 (x_2 - x_y t) + c_3 e_3 (x_3 - x_y t) \right) \quad (18)$$

$$x_y^{(t+1)} = x_y t + v_y^{(t+1)} \quad (19)$$

In the HPSGWO optimization technique, the fitness function is calculated as:

$$f(x) = \sum_{y=1}^{N-1} \left[100 \left(x_{(y+1)} - x_y^2 \right)^2 + (1 - x_y)^2 \right] \quad (20)$$

In order to classify different kinds of brain tumors using MRI images, this research proposes three completely autonomous CNN models. The CNN models' primary hyperparameters are automatically

tweaked using an HPSGWO. The first of these CNN models can predict whether a patient's tumor is visible on an MRI image, making it a valuable tool in the fight against brain cancer. The architecture designated Classification_1 used throughout this study is seen in Figure 4. Since it is intended to categorize images into two groups, the output layer consists of just two neurons. The softmax classifier uses the output of the last fully connected layer to make its final prediction. Table 2 provides further information on the CNN structure (Classification_1).

Algorithm: Hybrid Particle Swarm Gray Wolf Optimization (HPSGWO)

Initialize Gray Wolf population. X_i ($i = 1, 2, \dots, n$)

Initialize A , a , C and w

Randomly Initialize an agent of n wolves positions $\in [0, 1]$

Based on the fitness function attain the α, β, γ solutions.

Evaluate the fitness of agents by using Equation ()

While ($t < \text{Max_iter}$)

For each population

Update the velocity using Equation ()

Update the position of agents using Equation ()

end

Update A , a , C and w

Evaluate all particles using the objective function

Update the positions of the three best agents α, β, γ

$t = t + 1$

end While

FIGURE 3 HPSGWO algorithm.

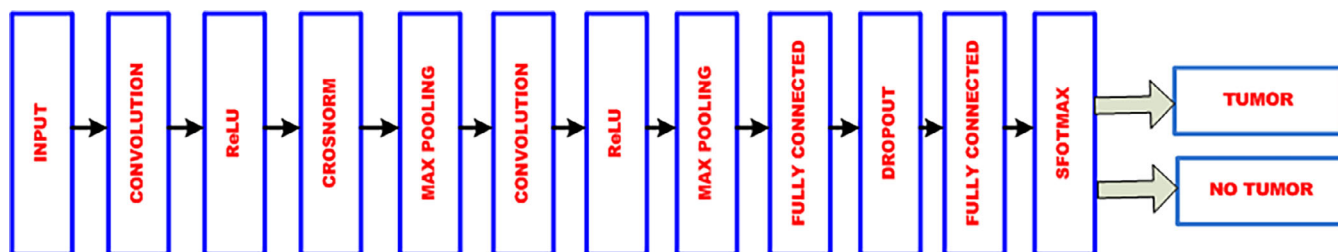


FIGURE 4 Architecture of Classification-1 work.

The second CNN model distinguishes between normal, glioma, meningioma, pituitary, and metastatic brain tumors. Classification_2 is the model's name throughout the study, and Figure 5 depicts its structure. The output layer consists of five neurons since it was trained to classify input images into those five classes. The softmax classifier predicts tumor type based on input from the last fully connected layer, which is a five-dimensional feature vector. Table 3 provides further information on the CNN structure (Classification_2).

TABLE 2 Optimum hyperparameters tuned by Hybrid Particle Swarm Grey Wolf Optimization for the Classification_1 work.

Parameter	Parameter ranges	Optimal value
Maximum pooling and convolution layers count	[1–4]	6
FC layer count	[1–4]	2
Filter count	[16, 24, 32, 48, 64, 96, 128]	128, 96
Size of the filter	[3–7]	6, 6
Activation function	[ELU, SELU, ReLU, Leaky ReLU]	ReLU
Mini-batch size	[4, 8, 16, 32, 64]	32
Momentum	[0.80, 0.85, 0.9, 0.95]	0.85
Learning rate	[0.0001, 0.0005, 0.001, 0.005]	0.001
l_2 Regularization	[0.0001, 0.0005, 0.001, 0.005]	0.0001

The third CNN model classifies grades II, III, and IV as glioma brain tumors. Classification_3 will be used to refer to this work for the rest of the study, and Figure 6 will depict the building's layout. Due to its design for segmenting images into three categories, the output layer consists of only three neurons. The softmax classifier, which makes the ultimate judgment on tumor grade, is fed a three-dimensional feature vector from the last fully connected layer. Table 4 provides further information on the CNN structure (Classification_3).

TABLE 3 Optimum hyperparameters tuned by Hybrid Particle Swarm Grey Wolf Optimization for the Classification_2 work.

Parameter	Parameter ranges	Optimal value
Maximum pooling and convolution layers count	[1–4]	6
FC layer count	[1–4]	2
Filter count	[16, 24, 32, 48, 64, 96, 128]	128, 96, 96, 24, 24, 32
Size of the filter	[3–7]	6, 6, 2, 6, 6, 4
Activation function	[ELU, SELU, ReLU, Leaky ReLU]	ReLU
Mini-batch size	[4, 8, 16, 32, 64]	64
Momentum	[0.80, 0.85, 0.9, 0.95]	0.85
Learning rate	[0.0001, 0.0005, 0.001, 0.005]	0.0001
l_2 Regularization	[0.0001, 0.0005, 0.001, 0.005]	0.001

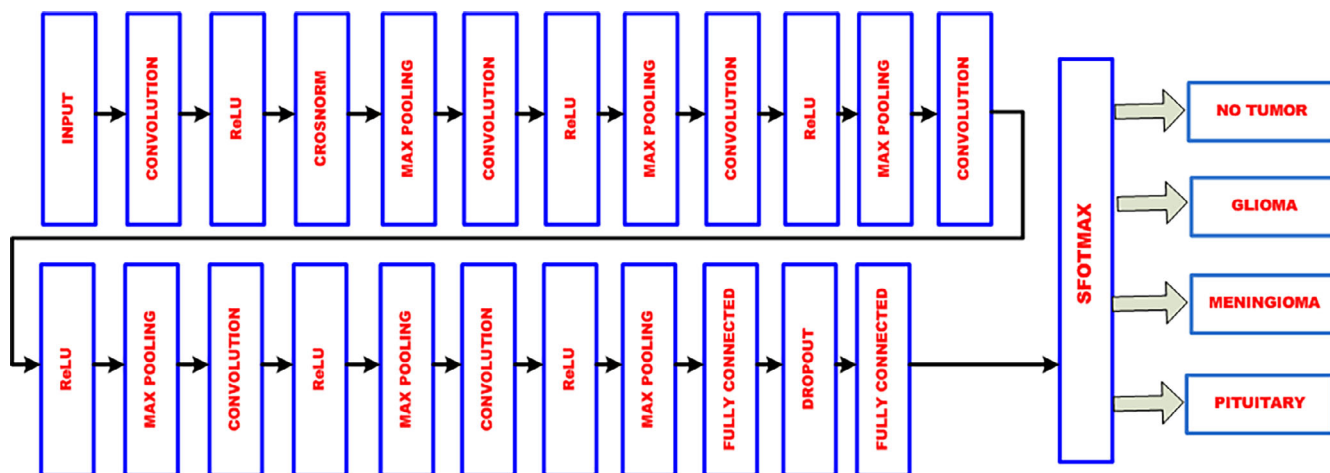


FIGURE 5 Architecture of Classification-2 work.

3.3 | Performance evaluation metrics

For scientific credibility, it is essential to evaluate the accuracy rate of categorization in image classification investigations. If so, the study's effort at classification is weak. Since the early days of image classification research, several kinds of performance evaluation criteria have been used and are now widely accepted as the norm. Accuracy, sensitivity, and specificity are all examples of such characteristics. These metrics are widely used as performance assessment criteria in studies of image classification, and they are also used in this work to assess the classification process's accuracy and dependability.

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (21)$$

$$\text{Sensitivity} = \frac{TP}{TN + FN} \quad (22)$$

$$\text{Precision} = \frac{TP}{TN + FP} \quad (23)$$

$$\text{Accuracy} = \frac{TP + TN}{TN + FP + FN + TP} \quad (24)$$

where TP, TN, FP, and FN stand for True Positive, True Negative, False Positive, and False Negative, respectively.

4 | EXPERIMENTAL RESULTS

Here, we report the results of tests performed with the supplied dataset and the models stated previously. Training and validation findings with high accuracy are worthless if the trained and hyperparameter-optimized CNN is not tested on anticipating unknown data. To ensure that

the high accuracy is not the result of biased dataset assignment, a test dataset is randomly generated from the training datasets and used to evaluate the trained CNN's ability to predict samples.

An excellent technique to see the traits the CNN develops during training is to display the activations of its convolutional layers. This visualization will let the researcher examine the results of the network. Simple features, such as colors and edges, are trained in a CNN's first convolutional layer, while more complex features, such as the boundaries of brain tumors, are learned in the second. In order to construct features for the later (deeper) convolutional layers, information from the earlier (shallower) layers is combined.

TABLE 4 Optimum hyperparameters tuned by Hybrid Particle Swarm Grey Wolf Optimization for Classification_3 work.

Parameter	Parameter ranges	Optimal value
Maximum pooling and convolution layers count	[1–4]	3
FC layers count	[1–4]	2
Filter count	[16, 24, 32, 48, 64, 96, 128]	128, 96, 96
Size of the filter	[3–7]	6, 6, 2
Activation function	[ELU, SELU, ReLU, Leaky ReLU]	ReLU
Mini-batch size	[4, 8, 16, 32, 64]	32
Momentum	[0.80, 0.85, 0.9, 0.95]	0.85
Learning rate	[0.0001, 0.0005, 0.001, 0.005]	0.005
l_2 Regularization	[0.0001, 0.0005, 0.001, 0.005]	0.001

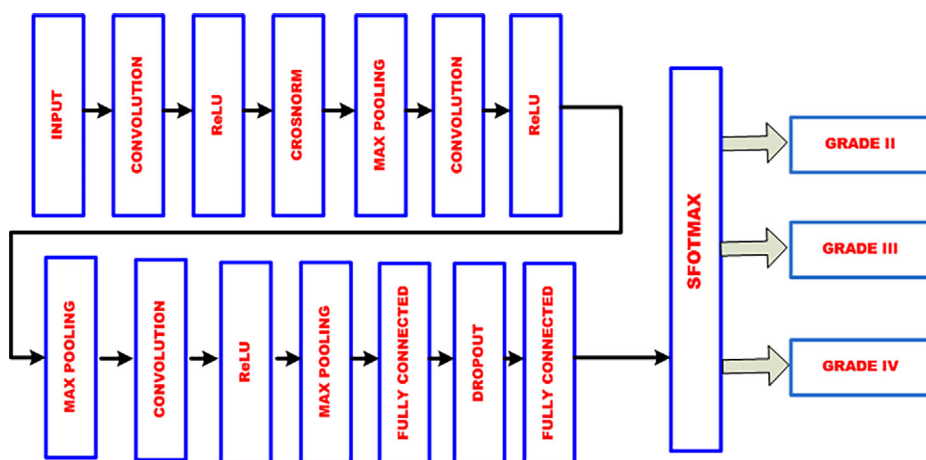


FIGURE 6 Architecture of Classification-3 work.

FIGURE 7 Activations of a given channel and its strongest counterpart of Classification_1 work.

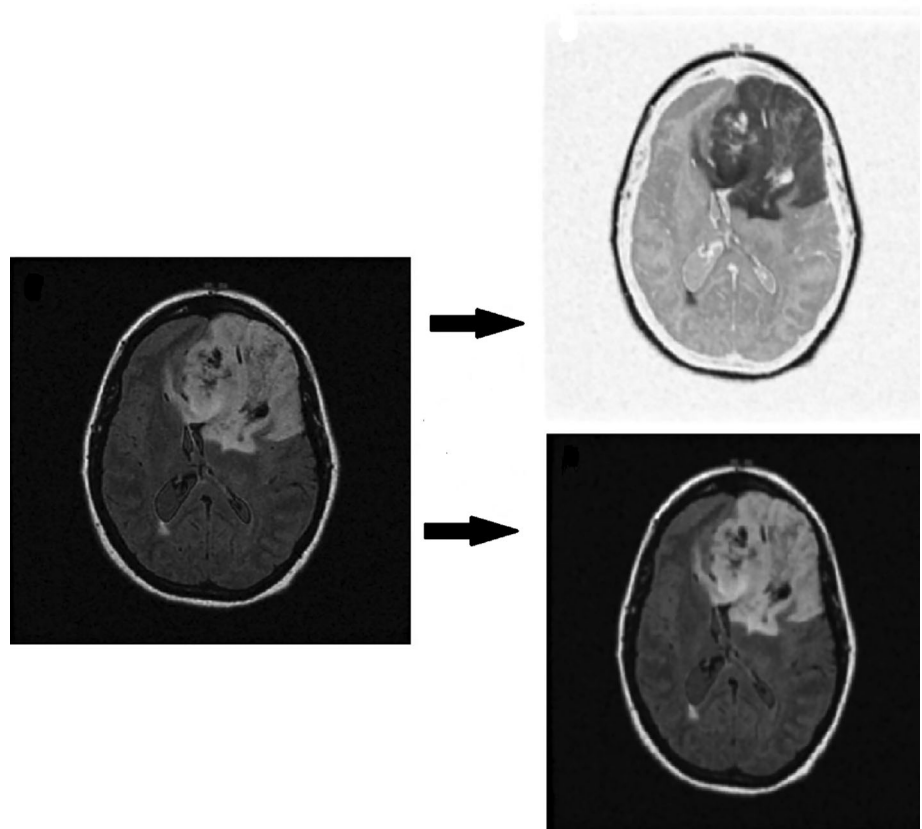


TABLE 5 Convolutional neural network model learning strategy.

Classification work name	Classification class name	Images count		
		Each class	Training (80%)	Testing (20%)
Classification_1	Tumor	2140	3192	798
	No Tumor	1850		
Classification_2	Normal	1350	5160	1290
	Glioma	1450		
	Meningioma	1200		
	Pituitary	1200		
	Metastatic	1250		
Classification_3	Grade II	2176	4856	1214
	Grade III	1718		
	Grade IV	2176		

CNN uses a 128-channel first convolutional layer and a 96-channel second convolutional layer for the Classification_1 work. Channels are two-dimensional arrays that make up each layer of a convolutional neural network. In Figure 7, the white channel pixels represent a channel that has been heavily activated by the tumor. CNN may have found tumors to be differentiating features in multiple image classes even though it was never told to do so. Unlike traditional artificial neural network techniques, which are often hand-developed for the

application at hand, these CNNs may learn to collect relevant characteristics on their own. In this study, learning to recognize tumors is useful for separating images with and without tumors. Table 5 displays the results of randomly dividing the 3990 samples from the research into training and test sets at a ratio of 80:20. Each training and testing set consists of 400 images, with 399 of those images randomly removed. The Classification_1 task model reaches 99.74% accuracy after 534 iterations. Figure 8 displays the classification results, while Figures 9

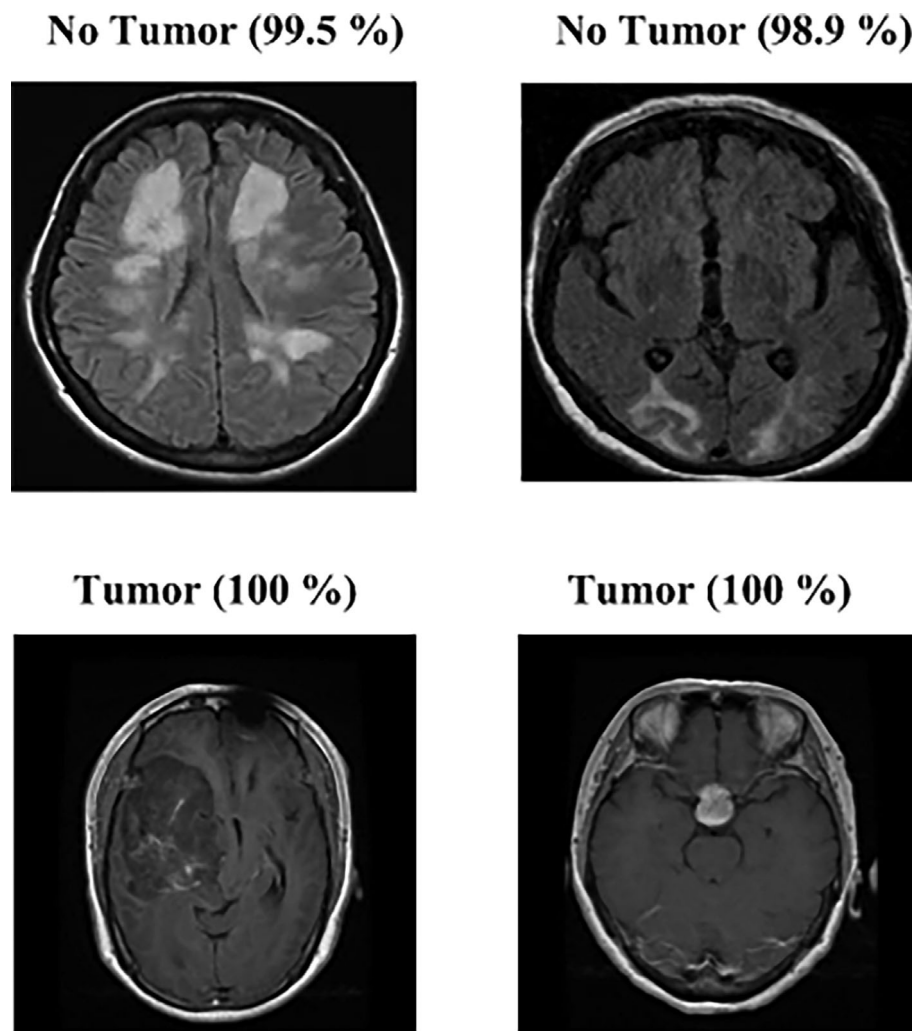


FIGURE 8 Classification results of Classification-1 work.

	No Tumor	Tumor
No Tumor	370	02
Tumor	00	426

FIGURE 9 Confusion matrix of Classification-1 work.

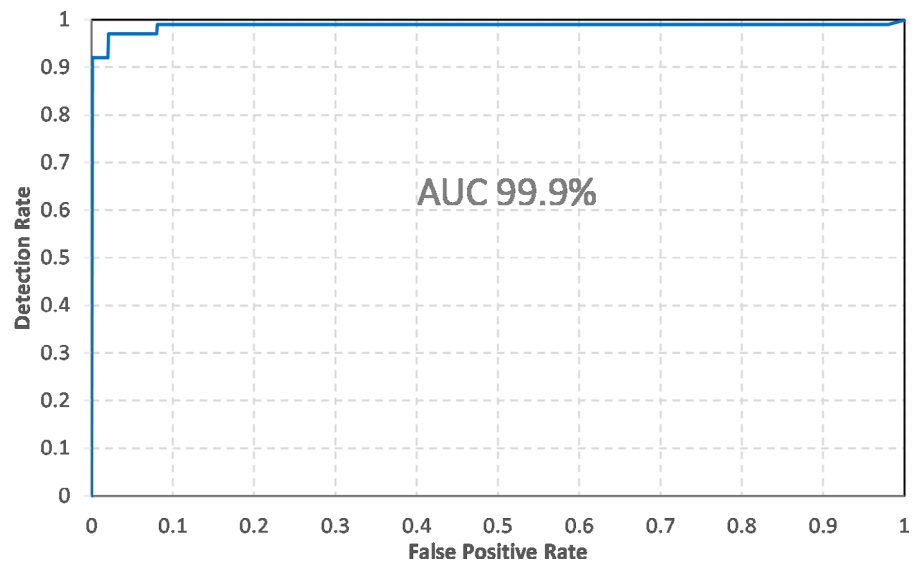
and 10 display the confusion matrix and receiver operating characteristic (ROC) curve analysis from Classification_1, respectively.

Table 5 displays that there are 6450 study samples, which is sufficient to randomly split into training and test

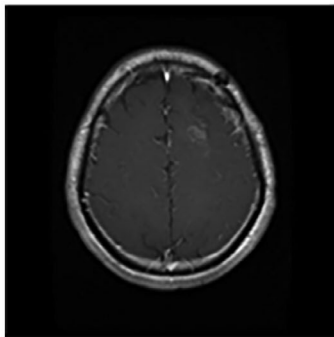
sets with a ratio of 80:20 for the Classification_2 problem. For the sake of model testing, 258 random images are removed from the datasets of each class. The proposed CNN model for the Classification_2 challenge achieves 96.27% accuracy in classification after 387 iterations. These results show that the proposed CNN model can distinguish between different varieties of brain tumors. The Classification_2 work's classification outcomes and confusion matrix are depicted in Figures 11 and 12, respectively. The ROC curve analysis of the Classification_2 is shown in Figure 13.

Table 5 shows that there are enough images for a training set and a test set with a ratio of 80:20 for the Classification_3 study's 6070 samples. A total of 404 random photos have been deleted from the dataset used to test the model for each class. The proposed model improves classification accuracy to 99.18% after 514 iterations. These findings provide credence to the suggested CNN technique for classifying brain tumors. Figure 14 displays the Classification-3 work's classification results, while Figures 15 and 16 display the work's confusion matrix and ROC curves analysis of the Classification_3.

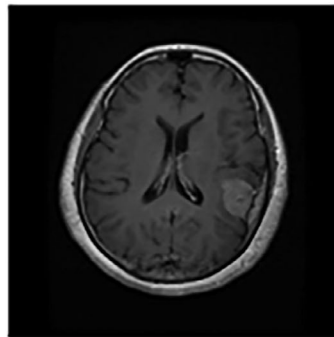
FIGURE 10 ROC curve analysis of Classification-1 work.



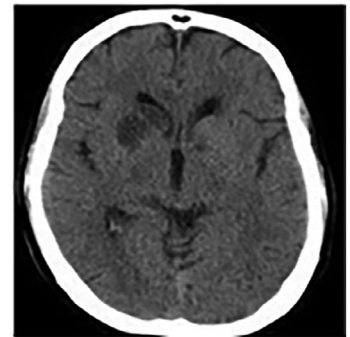
Glioma (99.18 %)



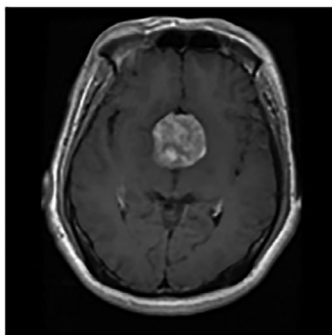
Meningioma (98.01 %)



No Tumor (100 %)



Pituitary (99.08 %)



Metastatic (98.31 %)

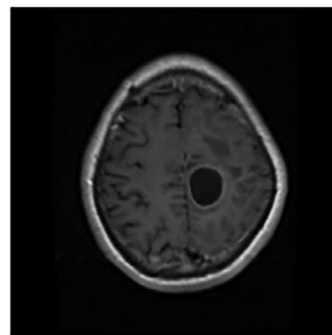


FIGURE 11 Classification results of Classification-2 work.

5 | COMPARISON AND DISCUSSION

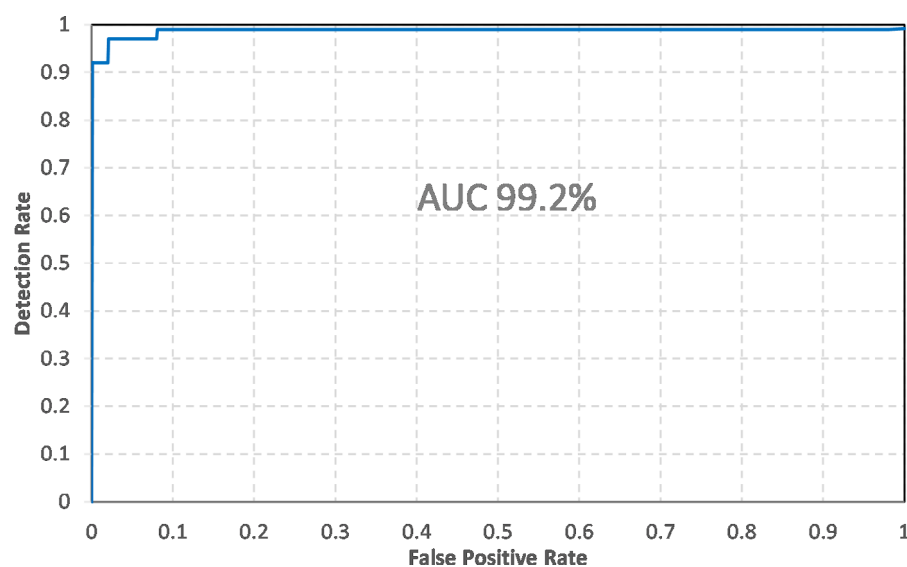
In this research, three distinct CNN models are created, one for each kind of categorization problem. The initial model's focus is on analyzing MRI images for signs of

brain tumors. Both the second and third models attempt to guess what kind of brain tumor it is and what stage it has progressed to. Convolutional neural networks encounter a number of issues, one of which is selecting the optimal network model for a given task. Particularly with CNNs, getting the optimal value of hyperparameters

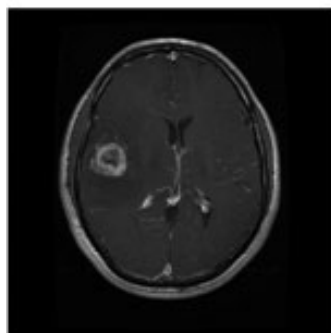
FIGURE 12 Confusion matrix of Classification-2 work.

	Glioma	Meningioma	Metastatic	Normal	Pituitary
Glioma	285	05	02	04	00
Meningioma	04	232	01	03	01
Metastatic	03	01	237	01	02
Normal	05	03	02	260	04
Pituitary	00	01	03	02	228

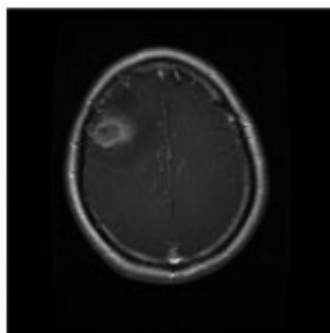
FIGURE 13 ROC curve analysis of Classification-2 work.



Grade II (99.36 %)



Grade III (100 %)



Grade IV (100 %)

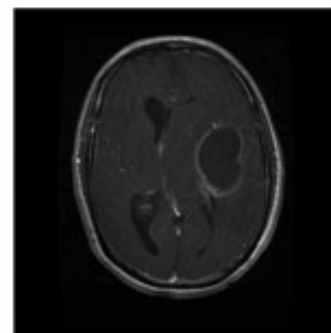


FIGURE 14 Classification results of Classification-3 work.

is a must for good outcomes. In this research, the HPSGWO optimizer is used to create the optimal CNN model and set its hyperparameters. Classification

	Grade II	Grade III	Grade IV
Grade II	431	00	07
Grade III	00	347	00
Grade IV	03	09	426

FIGURE 15 Confusion matrix of Classification-3 work.

findings are good when we are using large and freely available clinical datasets. In order to gauge how well the developed CNN models perform, it is instructive to compare their output to that of well-known pre-trained CNN models utilizing the same datasets, and the results are shown in Table 6.

For instance, the first designed CNN model can detect brain tumors with a highly acceptable accuracy of 99.74%. Also, 96.27% accuracy is achieved while classifying the different types of brain tumors. Last but not least, grading of the brain tumor was successful with high accuracy of 99.18%. Using performance evaluation criteria, the findings of the provided models are validated. Despite categorizing the tumor images into five categories, the suggested model for the Classification-2 test still achieves high accuracy. Obviously, the CNN models developed in this paper are better than existing techniques for multi-classifying brain tumor MRI images. In terms of performance evaluation and classification type, Table 7 compares the suggested models with the most recent methods described in the literature.

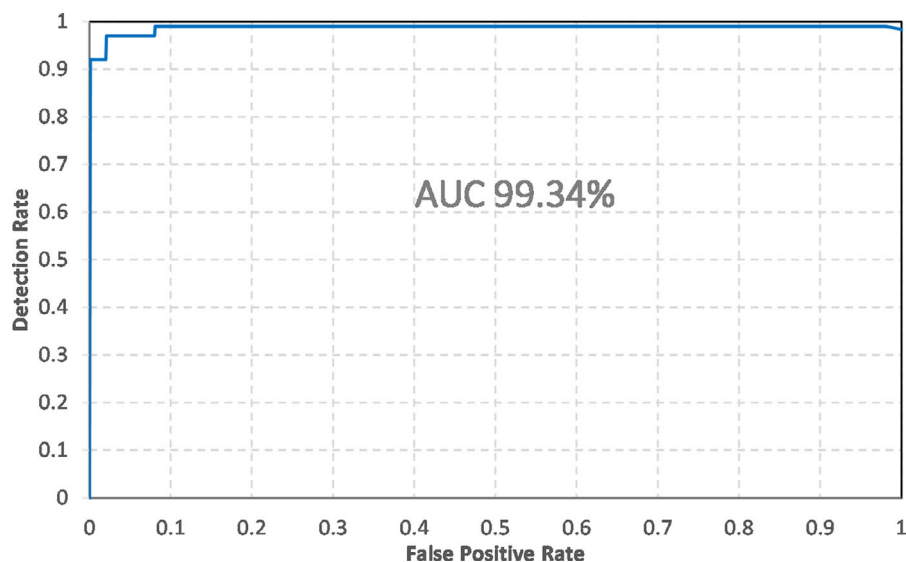


FIGURE 16 ROC curve analysis of Classification-3 work.

TABLE 6 Comparison summary of different pre-trained convolutional neural network models and proposed models.

CNN model	Accuracy		
	Classification_1	Classification_2	Classification_3
EfficientNet	87.31%	83.38%	90.51%
AlexNet	89.87%	85.04%	93.92%
ResNet-50	96.84%	79.92%	91.62%
VGG-16	91.20%	87.72%	89.41%
GoogleNet	79.62%	80.96%	95.09%
Proposed method	99.74%	96.27%	99.18%

Note: Bold values are more superior than the other values.

TABLE 7 Comparison summary of different standard existing models and proposed models.

Methods	Year	Accuracy of Classification_1	Accuracy of Classification_2	Accuracy of Classification_3	Classification type	Classification method
Hossam et al. ¹²	2019		96.13%	98.7%	Multi	CNN
Thirumurugan et al. ¹³	2016	99.6%			Binary	ANFIS
Deepak et al. ¹	2019		97.1%		Multi	deep CNN-SVM
Ahmet et al. ²	2020	97.01%			Binary	Hybrid CNN
Xiaoqing et al. ⁹	2021		97.8%		Multi	CDLLC
Machiraju et al. ¹¹	2022		99.34%		Multi	Inception-v3 CNN
Emrah et al. ¹⁵	2021	99.33%	92.66%	98.14%	Multi	deep CNN
Simy et al. ²⁰	2022	98.01%			Binary	LSFHS
Rehna et al. ¹⁴	2023	98.53%			Binary	Adaptive-ANFIS
Meenakshi et al. ³²	2023	94.6%			Binary	GBKS GC-SVM
Sushreeta et al. ³³	2023	99.5%			Binary	EfficientNet
Alok et al. ³⁴	2023		98.5%		Multi	AlexNet CNN
Prince et al. ³⁵	2023	98.93%			Binary	VGG-16 CNN
Saravanan et al. ³⁶	2023		98.17%		Multi	EFCMC
Basant et al. ³⁷	2023		98.86%		Multi	BTC-fCNN
Hossein et al. ³⁸	2023	98.69%			Binary	IACO-ResNet
Aparajita et al. ³⁹	2023	96%			Binary	Saliency-K-mean-SSO-RBNN
Chaosheng et al. ²³	2023			97.98%	Multi	GAM-SpCaNet
Muhammad et al. ²⁴	2023			96.72%	Multi	Saliency-Optimum CNN
Sarah et al. ²⁵	2023		95.69%		Multi	HHOCNN
Taha et al. ²⁶	2023		96.13%		Multi	PatchResNet
Rajkumar et al. ²⁷	2023		94%		Multi	PSPNet
Maibam et al. ⁴⁰	2023			98.3%	Multi	CNN-YOLOv3
Proposed method		99.74%	96.27%	99.18%	Multi	Deep CNN—HPSGWO

Note: Bold values are more superior than the other values.

In the comparison of the proposed model with the different existing models and pre-trained CNN models, we have used the same datasets which we have used in our model. In most of the cases, we have observed that there is a maximum of only two classification works, namely detection of the tumor and the classification of different types of tumor. But our model has presented three different classification works which are to detect tumors, classify different types of tumors, and the severity of the brain tumors of the types. It is clearly shown in Tables 6 and 7 that our proposed model is superior to the existing state-of-the-art and

pre-trained model in terms of accuracy. The ROC curves analysis clearly shows the accurate medical diagnosis of our model in all the different classifications works.

Our model will help in the early detection and speedy diagnosis of brain tumors since it will help doctors and radiologists to provide accurate results. They can compare the automated results with their results so that it will accurately diagnose the exact type of the tumor with the severity of the patients.

Authors has conducted a thorough literature review and have come to the following conclusions on the

benefits and contributions of the methodology provided in this research:

- CNN has never before been used to do multi-category classification on MRI images of brain tumors, hence this research is groundbreaking. Nearly all of CNN's hyperparameters are selected mechanically by the HPSGWO optimizer.
- With the use of the HPSGWO method, the ideal hyperparameters for the chosen CNN model can be determined.
- The innovative CNN model proposed for Classifications_1 achieves a 99.74% accuracy to detect brain tumors.
- When applied to the classification of MRI images of glioma, meningioma, pituitary, metastatic tumor types, and healthy persons, the unique CNN model proposed in Classifications_2 work achieves an impressive accuracy of 96.27%.
- When applied to brain tumor images, the innovative CNN model of the Classifications_3 work achieved a 99.18% accuracy rate for identifying grades II, III, and IV tumors.

6 | CONCLUSIONS

Deep learning's recent advancements have pushed the field of machine learning beyond feature engineering and into the realm of architectural engineering. In this research, CNN models are used to multi-classify brain tumors for early diagnosis, with almost all of the hyperparameters automatically tweaked using HPSGWO. Using publicly accessible medical image datasets, this work introduces three reliable CNN models for brain tumor classification. The detection rate for malignant brain tumors is 99.74%. In addition, a satisfying 96.27% accuracy is reached when sorting brain MRI images into distinct categories such as glioma, meningioma, pituitary, normal brain, and metastatic. There is a 99.18% accuracy rate when classifying glioma brain tumors as grades II, III, or IV. The proposed CNN models are trained and tested on a substantial dataset of medical images. The results of the proposed CNN models are compared with the standard existing methods of brain tumor classification and it clearly shows the superiority. The CNN models created for this research can be used by doctors and radiologists to verify their first screening for brain tumor classification.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in The Cancer Imaging Archive (TCIA) <https://doi.org/10.1007/s1> at <https://wiki.cancerimagingarchive.net/display/Public/RIDER+NEURO+MRI#225127532d56225fc49c45148bd9b89865c26ef2>, reference number <https://doi.org/10.1007/s10278-013-9622-7>.

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